

WORKFLOW VS AGENT DESIGN

Module 3 — Governed AI Feature Delivery

THE PROBLEM WE'RE SOLVING

Teams overuse agents where deterministic workflows would be safer.

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Now: apply a practical decision framework.

WHY TEAMS GET THIS WRONG

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- Execution boundaries are not defined before implementation.
- Tool usage is left too open-ended.
- Failure and traceability costs are discovered too late.

THREE EXECUTION PATTERNS

Pattern	Best for	Trade-off
Deterministic workflow	Repeatable, auditable tasks	Less flexibility
Bounded tools	Controlled external actions	Added orchestration complexity
Agentic flow	Open-ended planning and exploration	Harder governance and testing

PATTERN 1: DETERMINISTIC WORKFLOW

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- Known tool set and output contract
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- Best for extraction, classification, routing, compliance
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PATTERN 1: DETERMINISTIC WORKFLOW

- Fixed sequence with explicit branching
- Known tool set and output contract
- Best for extraction, classification, routing, compliance
- Easiest to test, evaluate, and audit

PATTERN 2: BOUNDED TOOL USAGE

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- Tools are permitted but within strict boundaries
- Tool contract defined by schema and policy
- Good when one or two external enrichments are required
- Preserves control while extending capability

PATTERN 3: AGENTIC BEHAVIOUR

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- Adaptive planning and dynamic decision chains
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Agentic is not the advanced default, it is the high-cost option.

DECISION FRAMEWORK

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- Need deterministic output? Prefer workflow.
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DECISION FRAMEWORK

- Need deterministic output? Prefer workflow.
- Need bounded external actions? Use restricted tools.
- Need adaptive multi-step exploration? Consider agent.
- If uncertain, start deterministic and evolve with evidence.

PRACTICAL DECISION QUESTIONS

- 1.
- 2.
- 3.
- 4.
- 5.

PRACTICAL DECISION QUESTIONS

1. Is deterministic behaviour required?
- 2.
- 3.
- 4.
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PRACTICAL DECISION QUESTIONS

1. Is deterministic behaviour required?
2. Is strict auditability required?
- 3.
- 4.
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PRACTICAL DECISION QUESTIONS

1. Is deterministic behaviour required?
2. Is strict auditability required?
3. How many tools are truly needed?
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PRACTICAL DECISION QUESTIONS

1. Is deterministic behaviour required?
2. Is strict auditability required?
3. How many tools are truly needed?
4. Can quality be evaluated repeatably?
5. What is the blast radius if tool calls go wrong?

Answer these before implementation not after.

WORKFLOW VS AGENT: PREDICTABILITY AND AUDITABILITY

Dimension	Workflow	Agent
Predictability	High	Medium / Low
Auditability	High	Medium

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Lower predictability means harder release criteria and more evaluation overhead.

WORKFLOW VS AGENT: TOOLING AND TESTABILITY

Dimension	Workflow	Agent
Tool control	Explicit	Dynamic
Testability	Straightforward	Harder
Failure analysis	Easier	More complex

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Dynamic tool selection is a governance cost, not just an implementation detail.

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- Constrain parameters and payload sizes explicitly.
- Define timeouts and retry behaviour per tool.
- Log tool call intent and outcome in every trace.

MCP TOOLS VS INTERNAL TOOLS

	MCP tools	Internal / master tools
Best for	Standard integrations, broad capability	Policy-sensitive, domain-specific operations
Ownership	External or platform-managed	Team-owned with tighter controls
Governance	Explicit boundaries required	Explicit boundaries required

The governance obligation is identical, the ownership model is not.

CASE STUDY A: DOCUMENT EXTRACTION

Need: parse known fields from known document classes

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CASE STUDY A: DOCUMENT EXTRACTION

Need: parse known fields from known document classes

- Deterministic workflow is almost always the right pattern
- Bounded tools optional for enrichment steps
- Agentic pattern introduces unnecessary risk and overhead

CASE STUDY B: EMAIL TRIAGE AND ROUTING

Need: classify intent, route to teams, enrich context

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CASE STUDY B: EMAIL TRIAGE AND ROUTING

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- Workflow plus bounded tools is usually the strongest pattern
- Agent behaviour may be justified for ambiguous edge cases only
- Route uncertainty should trigger fallback to human review

CASE STUDY C: OPEN-ENDED INVESTIGATION ASSISTANT

Need: explore documents and systems, propose next actions

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CASE STUDY C: OPEN-ENDED INVESTIGATION ASSISTANT

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- Agentic behaviour may be justified here
- Requires deeper trace coverage and stronger guardrails
- Use approval checkpoints before high-risk actions execute

MIGRATION STRATEGY: WORKFLOW FIRST

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- Start with a deterministic baseline.
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MIGRATION STRATEGY: WORKFLOW FIRST

- Start with a deterministic baseline.
- Measure where flexibility is actually needed: use evidence, not assumption.
- Introduce bounded tools before moving to autonomy.
- Add agentic behaviour only when evaluation results support it.

FAILURE MODES TO WATCH

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- Over-agentic design applied to deterministic problems
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FAILURE MODES TO WATCH

- Over-agentic design applied to deterministic problems
- Too many tools with weak parameter contracts
- No trace path covering dynamic tool decisions
- No fallback plan when confidence is low or tool calls fail

WHAT WE PRODUCE IN MODULE 3

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WHAT WE PRODUCE IN MODULE 3

- Execution pattern decisions for your key use cases
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WHAT WE PRODUCE IN MODULE 3

- Execution pattern decisions for your key use cases
- Tool boundary rules and parameter constraints
- A reusable decision checklist for your team
- Rationale to carry directly into the security and guardrails module

SUMMARY

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1. **Agentic is not the default:** choose autonomy only when justified.
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2. **Workflow first** keeps delivery predictable and auditable.
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SUMMARY

1. **Agentic is not the default:** choose autonomy only when justified.
2. **Workflow first** keeps delivery predictable and auditable.
3. **Bounded tools** offer a practical and controllable middle ground.
4. **Decision frameworks** replace intuition with defensible engineering rationale.

BRIDGE TO MODULE 4

What we've established:

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What's next:

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BRIDGE TO MODULE 4

What we've established:

- Execution model decisions with explicit rationale.
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- Tool boundaries that define your attack surface.

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What's next:

- Regardless of execution model, the same attack vectors apply.
- Your first task in Module 4: map your tool boundaries against the threat model.

Module 4 covers security and guardrails — securing the end-to-end AI feature path.

QUESTIONS?

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